
Test-Time Adaptation with TENT Across Multiple Benchmarks

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Abstract

Deep neural networks perform well on standard benchmarks but degrade sharply under distribution shift. Test-time adaptation (TTA) addresses this problem by updating models online using only incoming test data. This project conducts a systematic evaluation of TENT (Test-time Entropy minimization) across CIFAR-10-C, CIFAR-100-C, and ImageNet-C, examining both single-corruption and sequential corruption adaptation. The results show that TENT consistently improves robustness, often within the first batches of adaptation. The experiments further reveal clear differences in adaptation behavior across dataset scale, corruption families, and sequential shift conditions.

1 Introduction

Deep neural networks assume that training and test data share the same distribution, but real-world inputs frequently violate this assumption due to noise, blur, lighting changes, weather, and sensor degradation. Even small shifts can substantially reduce accuracy.

Most prior robustness methods, like augmentation, adversarial training, or specialized architectures, operate only during training and cannot react to new shifts at deployment. Test-time adaptation (TTA) addresses this gap by updating the model online, without labels, as new data arrives.

TENT (Wang et al., 2021) is one of the simplest and most effective TTA methods to date. It adapts only BatchNorm affine parameters and minimizes prediction entropy on unlabeled test samples, yielding efficient and stable improvements. This project systematically evaluates TENT across:

- Single-corruption adaptation on CIFAR-10-C, CIFAR-100-C, and ImageNet-C
- Sequential corruption adaptation on CIFAR-10-C and CIFAR-100-C
- A step-ablation analysis on CIFAR-100-C

This work contributes to an empirical comparison across three benchmarks, an analysis of sequential adaptation stability, and a step-ablation study quantifying how quickly TENT converges.

2 Background

2.1 Distribution shift

Distribution shift occurs when the test distribution deviates from the training distribution. Hendrycks & Dietterich introduced the “Common Corruptions” benchmarks, showing that even strong models suffer large accuracy drops across noise, blur, digital, and weather corruptions. Their work highlighted

that modern neural networks often generalize poorly outside the training domain, motivating a large line of research on robustness and online adaptation.

2.2 TENT and other TTA methods

Early methods such as TTT (Sun et al., 2020) used auxiliary tasks, while later work like EATA (Niu et al., 2022) introduced regularization to prevent overfitting. These methods demonstrate that even lightweight inference-time updates can improve robustness.

In this context, TENT (Test-time Entropy minimization), proposed by Wang et al. (2021), is one of the simplest TTA methods. TENT adapts a model by minimizing prediction entropy: $H(p) = -\sum_c p_c \log(p_c)$. It updates only BatchNorm affine parameters (scale, shift). This restricts adaptation to normalization statistics, preventing large drifts in convolutional weights and enabling fast, stable inference-time updates. Although simple, TENT has become a standard baseline for TTA research because it is model-agnostic, requires no architectural changes, and performs reliably across corruptions, while simultaneously avoiding the instability and computational overhead associated with methods that update full network weights.

3 Methodology

All experiments follow the same pipeline:

1. Load a pretrained model
2. Enable only BatchNorm affine parameters for optimization
3. Evaluate initial baseline accuracy
4. Run TENT batch-by-batch
5. Evaluate adapted accuracy

3.1 Models and baseline training

CIFAR experiments use a ResNet-18 pretrained on ImageNet and a final layer fine-tuned on CIFAR-10/100. ImageNet-C uses a standard ResNet-50, already pretrained on ImageNet. All weights except BN affine parameters are frozen during TENT.

3.2 Single-corruption adaptation

For all single-corruption experiments, I evaluate the following four corruptions: Gaussian noise, motion blur, brightness, and fog. These corruptions cover the most prominent corruption types encountered in the real world (noise, blur, digital, weather). TENT was run for 400 steps on CIFAR-10-C, 300 steps on CIFAR-100-C, and 200 steps on ImageNet-C, balancing computational cost with the diminishing returns observed in section 4.4.

3.3 Sequential domain adaptation

To test TENT in non-stationary environments, I evaluated using three corruption sequences:

- Sequence A, winter weather: frost $\rightarrow$ snow $\rightarrow$ fog
- Sequence B, sunny to rainy: brightness $\rightarrow$ fog $\rightarrow$ spatter
- Sequence C, camera degradation: defocus blur $\rightarrow$ Gaussian noise $\rightarrow$ jpeg compression

For each corruption, CIFAR-10-C uses 400 adaptation batches, while CIFAR-100-C uses 200. These experiments measure how well BN-only adaptation carries across changing domains. Because each corruption induces a different shift in feature statistics, these sequences test whether TENT can preserve useful information across environment transitions or whether its BN-specific updates overfit to each local domain.

3.4 Step ablation procedure

To better understand TENT’s adaptation dynamics and to determine an efficient number of test-time optimization steps, I conducted a step ablation study using CIFAR-100-C, batches of size 128, and 300 total steps. The goal is to characterize how accuracy improves (or saturates) as TENT is run for increasing numbers of adaptation batches. I selected three representative corruption types: Gaussian noise, motion blur, and fog, since they cover high-frequency noise, low-frequency blur, and weather-related distribution shifts.

4 Results

4.1 Single-corruption results

I first examine TENT’s effect on CIFAR-10-C, CIFAR-100-C, and ImageNet-C when adapting on one corruption type at a time. Across all datasets, TENT consistently improved accuracy relative to the non-adapted baseline. The magnitude of improvement depended on dataset scale and corruption type.

- CIFAR-10-C had the largest relative accuracy gains (about 30%), consistent with prior work in small-scale TTA
- CIFAR-100-C gains were more modest (about 10%), reflecting the increased difficulty
- ImageNet-C still showed measurable improvements, but gains varied substantially by corruption

Corruption	Initial Acc	Final Acc	Improvement	Initial Acc	Final Acc	Improvement	Initial Acc	Final Acc	Improvement
gaussian noise	47.5	69.7	22.2	33.8	47.2	13.4	47.4	44.5	-2.9
motion blur	33.9	71.5	37.6	39.1	50.3	11.2	31.6	52.1	20.5
brightness	52.5	74.7	22.2	51.3	53.5	2.2	70.3	75.7	5.4
fog	32.3	68.4	36.1	36.9	46.2	9.3	37.6	64.5	26.9

Figure 1: Single-corruption accuracy for CIFAR-10-C (left), CIFAR-100-C (middle), and ImageNet-C (right).

4.2 Sequential corruption results

I next evaluate TENT under a sequential stream of corruptions, where the model adapts continually as the corruption type changes. I performed this experiment on both CIFAR-10-C and CIFAR-100-C. Results demonstrate that accuracy rises within each corruption segment and drops sharply at the corruption boundaries. While the sequential adaptation on CIFAR-10-C nearly matched its single-corruption performance, the accuracies on CIFAR-100-C ended about 2-8% lower than their single-corruption counterparts.

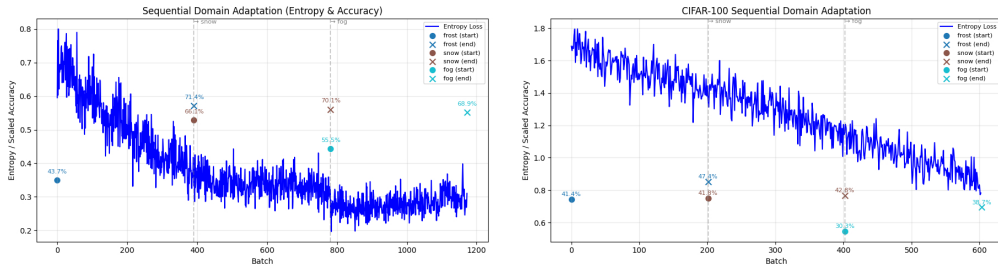


Figure 2: Sequential corruption domain adaptation results: CIFAR-10-C (left) and CIFAR-100-C (right).

4.3 Analysis of entropy trends

- Entropy decreases steadily within each corruption
- Accuracy often plateaus before entropy does
- Larger class counts lead to higher overall entropy and smaller decreases during TENT

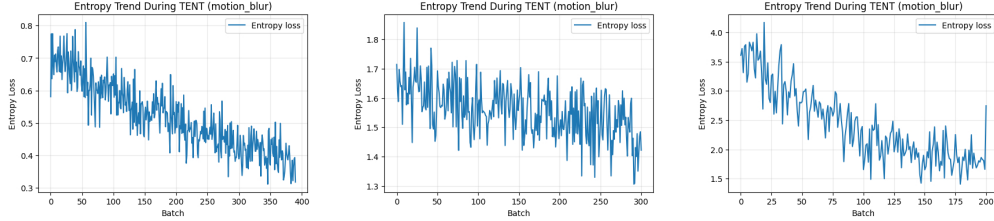


Figure 3: Entropy trend during TENT for motion blur. CIFAR-10-C (left), CIFAR-100-C (middle), and ImageNet-C (right).

4.4 Ablation on steps results

	Initial	5	10	20	40	80	160	300
gaussian noise	33.8	42	44.9	45.9	45.94	46.4	47.1	47.7
motion blur	39.1	44.5	47.1	49.2	49.7	49.8	50.4	50.4
fog	33.8	47.5	47.6	47.6	47.7	47.8	47.7	47.5

Figure 4: Accuracy at different step counts for three corruption types on CIFAR-100-C.

Most improvements occur within the first 5–20 steps. Beyond about 40 steps, returns diminish, and long schedules can introduce overfitting.

5 Conclusion

5.1 Overall performance and dataset complexity

Across all experiments, TENT consistently improved model performance under distribution shift, validating entropy minimization as a simple but powerful method for test-time adaptation. The consistency of improvement strongly depends on dataset complexity: gains were large on CIFAR-10-C, moderate but reliable on CIFAR-100-C, and quite variable on ImageNet-C.

5.2 Corruption-dependent behavior

TENT’s effectiveness varied significantly across corruption types. Weather- and blur-based corruptions produced the most stable and substantial improvements, likely because these shifts induce strong, systematic changes in BatchNorm statistics that TENT can correct. High-frequency noise corruptions (e.g., Gaussian noise) yielded smaller gains, as their unpredictable, aliasing-heavy distortions degrade feature extraction itself rather than merely shifting activation statistics. Digital corruptions such as brightness saw modest improvements but already-high baseline accuracy, reflecting an upper limit on what unlabeled entropy minimization can correct. This suggests that TENT is most effective when the corruption causes a coherent shift in activation statistics rather than destructive perturbations that impair feature extraction.

5.3 Sequential adaptation limitations

The sequential corruption experiments reveal a key limitation of continuous test-time adaptation. Although TENT quickly adapts to each corruption, those adapted BatchNorm parameters do not fully transfer to the next domain. This produces a repeating pattern: accuracy rises during each corruption segment and drops sharply at corruption boundaries. This exposes a fundamental trade-off in continuous adaptation:

- Insufficient adaptation → poor performance within the current corruption
- Accuracy often plateaus before entropy does

CIFAR-10-C handled this well, with sequential performance nearly matching its single-corruption results. CIFAR-100-C, however, experienced a 2–8% drop, likely because its larger label space makes TENT’s confidence-based updates less informative, amplifying drift when the domain shifts. These findings support the motivation behind more recent methods such as EATA (Niu et al., 2022), which introduces explicit regularization to prevent drift.

5.4 Entropy-accuracy relationship

Entropy consistently decreased during TENT adaptation, even in cases where accuracy plateaued. This highlights an important principle: TENT optimizes confidence, not correctness.

Lower entropy corresponds to more confident predictions, but not necessarily more accurate ones. This explains why entropy continues falling while accuracy stagnates, especially in high-class or noise-heavy regimes. The step-ablation study reinforces this: the majority of accuracy gains occur early (within 5–20 steps), while additional entropy reduction beyond about 40 steps contributes little and may introduce overfitting.

5.5 Summary

Overall, the experiments indicate that TENT is:

- Simple and stable – BN-only updates avoid catastrophic forgetting
- Fast to converge – most benefits appear within just a few batches
- Highly effective on many corruptions – particularly blur and weather
- Best suited for lower complexity datasets – CIFAR-10-C > CIFAR-100-C ImageNet-C

However, TENT also exhibits limitations:

- Over-adaptation during sequential shifts – requires anti-drift regularization
- Inconsistent gains on large-class datasets – entropy minimization becomes less informative
- Long adaptation schedules yield diminishing returns

Taken together, these results position TENT as a strong, efficient, and reliable baseline for test-time adaptation. The findings also clarify when TENT should be preferred over heavier TTA methods: namely, in settings requiring rapid, low-overhead adaptation without modifying core model weights. At the same time, they highlight the need for more advanced methods that incorporate regularization, confidence filtering, or memory mechanisms to maintain stability under non-stationary test streams and large output spaces.

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